# A Multi-Agent Based Simulation Modeling Method for UAV Swarm Collaborative Combat


Linpeng Jia
*School of Automation*
*Beijing Institute of Technology*
*The National Key Lab of Autonomous Intelligent Unmanned Systems*
Beijing, China
13153534895@163.com

Lingda Wang
*School of Automation*
*Beijing Institute of Technology*
*The National Key Lab of Autonomous Intelligent Unmanned Systems*
Beijing, China
2593934150@qq.com

Wei Quan
*School of Automation*
*Beijing Institute of Technology*
*The National Key Lab of Autonomous Intelligent Unmanned Systems*
Beijing, China
weiquan_bit@hotmail.com

Enjun Qiao
*School of Automation*
*Beijing Institute of Technology*
*The National Key Lab of Autonomous Intelligent Unmanned Systems*
Beijing, China
956755388@qq.com

Zhao Zhang
*School of Automation*
*Beijing Institute of Technology*
*The National Key Lab of Autonomous Intelligent Unmanned Systems*
Beijing, China
bitzhangzhao@163.com

Chen Chen*
*School of Automation*
*Beijing Institute of Technology*
*The National Key Lab of Autonomous Intelligent Unmanned Systems*
Beijing, China
xiaofan@bit.edu.cn



2024 39th Youth Academic Annual Conference of Chinese Association of Automation (YAC) | 979-8-3503-7922-8/24/$31.00 ©2024 IEEE | DOI: 10.1109/YAC63405.2024.10598485



***Abstract*—Unmanned Aerial Vehicle (UAV) swarm collaborative combat is increasingly becoming a vital component in modern warfare systems. As an important part of swarm warfare research, the modeling and full process simulation systems for collaborative operations have become a challenging topic. Considering the characteristics of swarm combat, this paper constructs a simulation modeling design for UAV swarm collaborative combat simulations. Based on multi-agent modeling technology, UAV swarms are modeled as multiple intelligent agents with behavior models and strategy models. In the behavior model, the movement, formation, communication, detection, and strike behaviors of UAV swarm are fully considered. To further characterize the collaborative characteristics, emergency network formation strategies, collaborative search strategies based on stepwise predictive control, and collaborative strike strategies responsive to situational changes are designed in the strategy model. A simulation scenario is designed and simulated, which validates the feasibility and effectiveness of the simulation model. The impact of swarm size, task complexity, environmental interference， and strategic selection on combat effectiveness is verified by the simulation system. This simulation model can fully show the validation for collaborative operations and provide guidance for decision optimization.**



***Keywords—UAV Swarm, Collaborative Combat, Simulation Model.***


## I. INTRODUCTION

Along with the evolution of warfare, the operational units in modern warfare are progressively moving towards miniaturization and swarm organization. As unmanned system technology matures, UAV swarms are increasingly utilized in military operations for diverse roles such as aerial reconnaissance, electronic warfare, communication relay, and precision strikes. The swarm's collaborative capability significantly enhances overall combat effectiveness, distinguishing swarm operations from individual combat operations in terms of strategic value. Consequently, UAV swarm collaborative combat is gradually becoming an integral part of modern warfare systems.

Currently, research on UAV swarm collaborative combat has emerged as a focal point in the field of warfare, drawing extensive attention from both domestic and international militaries and research institutions. The simulation of UAV swarms is an important part of equipment demonstration, which is of great significance for tactical evaluation and verification of tactical assessment and validation of swarm combat strategies, testing the operational efficacy of weapon systems, and evaluating swarm-level strategic maneuvers.

In recent years, there has been relatively limited research on combat simulation systems of UAV swarm. Wen [1] designed an air combat simulation system specifically for assessing the effectiveness of weapon performance. Duan [2] established a simulation environment and agent model for the targeted scenario of helicopters attacking armored targets. Liu [3] designed a tactical-level air combat simulation system for intelligent decision-making. Qing [4] develops a fundamental combat scenario reflective of contemporary naval battlefield conditions and enhances combat decision-making through the application of algorithms within a simulation and deduction system. Nevertheless, existing swarm simulation models exhibit certain limitations, often confined to specific operational phases or focusing on the evaluation of the performance of particular equipment.

Aiming at simulating the complete process of UAV swarm collaborative combat, this study designs a multi-agent based simulation model. It integrates diverse UAV agents with multiple behavior models and collaborative strategic models. The experiments on a detailed combat simulation experiments validate the feasibility and effectiveness of the simulation model.

## II. SIMULATION MODELING FRAMEWORK

To facilitate the simulation experimentation of UAV swarm collaborative combat scenarios, this paper utilizes intelligent agents as carriers for strategic execution to construct the architectural design of the simulation model and the operational logic of the simulation platform.

### *A. Architecture Design of Simulation Model*

In this study, we have constructed a comprehensive system simulation model framework, comprising six modules: Agent Model, Strategy Library, User Interface, Scenario Designation, Simulation Engine, and Data Module. The Agent Model and Strategy Library form the core repository of the model,


*Corresponding Author


This work was supported by the National Natural Science Foundation of China under Grant 62273044, 62022015 and 62088101.



Authorized licensed use limited to: NANJING UNIVERSITY OF AERONAUTICS AND ASTRONAUTICS. Downloaded on January 05,2026 at 09:09:12 UTC from IEEE Xplore. Restrictions apply.

providing the essential entities and strategic options for the simulation. These components are controlled through the User Interface, which facilitates task management and scenario loading. The system's Simulation Engine drives the simulation process, continuously interacting with the Data Module to ensure the seamless operation of the entire simulation model. The architecture of the simulation model is depicted in Fig. 1.

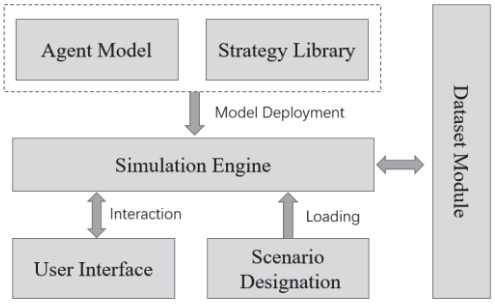



Fig. 1. Simulation Model Framework

### B. Simulation Run Logic

AnyLogic software is a multi-method simulation modeling software that enables users to create complex models and simulate a wide range of systems and processes, facilitating decision-making and problem-solving. It offers powerful simulation and optimization capabilities and finds applications across various domains, including manufacturing, logistics, healthcare, combat simulation, and more. Utilizing it, we have constructed the operational logic of the simulation platform with agents serving as carriers for strategy execution. In the simulation control module, a system simulation clock signal is established to build a discrete-time system simulation model. Within each clock signal interval, the scenario state update module, individual agent action modules, and performance verification module are sequentially executed. This ensures the real-time updating of each agent's behavioral decisions and task actions. A high-frequency clock signal is employed to simulate the conditions of continuous-time operation in the simulation.

## III. Construction of UAV Behavior Model

To address the unique movement characteristics of UAV in combat missions, we have designed specific models tailored to rotorcraft UAVs. These include the UAV Motion Model, Collaborative Behavior Model, Detection Model, and Strike Model. Each of these models encapsulates the distinct aspects of UAV operations, from their dynamic movement patterns to their coordinated activities, detection capabilities, and strike mechanisms in combat scenarios.

### A. Motion Model

The UAV model is constructed for rotary-wing UAVs. In three-dimensional space, the maneuvering state of the UAV can be represented using a state vector

$$Q = [P, V, H] \tag{1}$$

where $P = (x, y, z)$ is the current position vector of the UAV. $V = (v_x, v_y, v_z)$ is the current velocity vector of the UAV. $H = (h_h, h_p)$ is the direction vector for the UAV's current horizontal angle and pitch angle.

*1) Motion Control:* In the context of a discrete simulation framework, during each simulation clock cycle, the UAV undergoes a position update. It's based on the current UAV's velocity vector $V$, which is controlled by the scalar velocity $v$, and the direction vector $H$. Therefore, this paper designs a motion control model for the UAV based on speed control and orientation control. Six different UAV motion control actions are designed, including acceleration, deceleration, turning left, turning right, ascending, and descending. The UAV's behaviors such as free exploration, target-oriented movement, obstacle avoidance, and others are composed of combinations of these six control actions. This forms the basis of the UAV's motion control model in complex combat mission environments.

*2) Obstacle Avoidance:* In a battlefield environment, there are restrictions such as restricted areas and obstacles that the UAV cannot traverse. So the concept of artificial potential fields [5] is used to perform obstacle detection and avoidance behaviors. During the execution of a mission, when the direction vector $H$ detects an obstacle within a conical region in its forward direction, a virtual force is generated to guide changes in UAV's motion state, thereby achieving obstacle avoidance.

### B. Collaborative Behavior Model

*1) Formation Control:* The UAV, due to their individual capabilities, are categorized into three specialized types: Communication UAV, Detection UAV, and Strike UAV. They are responsible for inter-group communication, mission target detection, and engagement of mission targets, respectively. UAV swarms aggregate from a certain proportion of these three types of UAVs to form a task group, and the proportion of each task group can dynamically change during mission execution. For the formation configuration of each task group, multiple formation control shapes have been designed, including line formations, umbrella formations, cross formations, and polygons. A leader-follower formation control model has been developed. And multi-step desired position prediction is employed to guide individual motion control, ultimately achieving formation control to achieve formation control.

*2) Communication Networking:* In the context of swarm-based collaborative warfare, the swarm is divided into multiple task group units for mission execution. Each task group includes a communication UAV with an extended communication range, responsible for information exchange within the group and between different task groups. Throughout the collaborative warfare process, this enables functions such as target information sharing, area detection coverage, and emergency networking.

### C. Detection Model

The UAV detection model utilizes a truncated exponentially attenuated disk probabilistic sensing model [6], as described in reference. In contrast to the Boolean sensing model, the probabilistic sensing model takes into account that the coverage probability decreases monotonically with distance, making it a more applicable approach. Let's $P_i(j)$ denote the detection probability of point $j$ by sensor $i$. In

Authorized licensed use limited to: NANJING UNIVERSITY OF AERONAUTICS AND ASTRONAUTICS. Downloaded on January 05,2026 at 09:09:12 UTC from IEEE Xplore. Restrictions apply.

this context, a truncated exponentially attenuated disk probabilistic sensing model can be defined as

$$P_i(j)=\begin{cases}1 & \text{if } \mathrm{d}(ij)\le r_s\\ e^{-k\alpha^{\beta}} & \text{if } r_s<\mathrm{d}(ij)\le r_e\\ 0 & \text{if } \mathrm{d}(ij)>r_e\end{cases} \tag{2}$$

where $\alpha=d(i,j)-r_s$, and $k$, $\beta$ are attenuation factors.

### D. Strike Model

The strike UAV employs a strike model based on probabilistic engagement and damage threshold. Weapon modeling is based on missile-type weapons with fixed strike intervals, used for ammunition reloading, and other related processes. Due to environmental electromagnetic interference and the adversary's mobility, the weapon's hit probability on the target is not a deterministic value but follows a certain probability distribution. When the target's damage reaches a certain threshold after enduring a certain level of firepower.

## IV. Swarm Strategy Model Construction

Comprising multiple intelligent UAVs, a UAV swarm collaborates to conduct coordinated operations, leveraging mutual cooperation between individual UAVs to enhance the swarm's overall intelligence. To enable the swarm to execute complex tasks more efficiently, this section, built upon the UAV model, focuses on typical scenarios for collaborative operations within a UAV swarm. It designs a set of collaborative operational procedures and constructs corresponding models for swarm-based collaborative warfare strategies.

### A. Swarm Collaboration State

In this section, focused on typical scenarios for collaborative detection and strike warfare within a UAV swarm, a swarm state transition diagram is designed to enable state transitions and autonomous decision-making within the UAV swarm. The swarm determines its state through state perception and information exchange, allowing it to switch between different task phases. The swarm's mission states are primarily divided into the following four parts, as shown in the state transition diagram in Fig. 2.

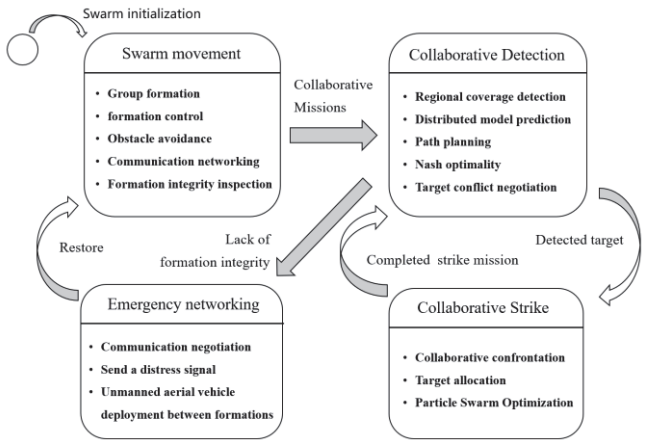



Fig. 2. Swarm State Transition Diagram

*1) Swarm Movement:* Swarm movement is carried out with task groups as the units, enabling formation flight. During this process, activities such as formation control, obstacle avoidance, communication networking, and formation integrity checks are conducted. The implementation primarily relies on the UAV behavior models described in the previous section.

*2) Emergency Networking:* The emergency networking state involves communication negotiations between swarms. Swarms that experience a loss of formation integrity will send distress signals to request assistance, enabling the reallocation of UAVs between swarms for coordination.

*3) Collaborative Detection:* The task during the collaborative detection phase is to conduct area coverage and detection. It involves designing a distributed model for prediction and communication-based target conflict negotiation to achieve a Nash equilibrium. This calculation results in the swarm's single-step path planning strategy, enabling the accomplishment of the collaborative detection mission.

*4) Collaborative Strike:* In the collaborative strike phase, the swarm operates in a formation to engage in coordinated combat against enemy UAVs, including the allocation of weapons for target engagement.

### B. Emergency Networking Strategy

During the execution of collaborative detection and combat missions, if factors such as combat damage or communication interference lead to an imbalance in the proportion of the three types of UAVs within a task group, the emergency networking procedure will be initiated. Communication UAVs within this group will broadcast a distress signal within their communication range, requesting assistance from UAVs in other groups.

When a group that meets the rescue criteria receives the signal, it undergoes a process of status analysis and negotiation. Subsequently, it dispatches an appropriate number of UAV swarms to assist, thereby maintaining the operational capability and resource balance among the various groups within the swarm.

### C. Collaborative Detection Strategy

Facing the task of collaborative detection of unknown areas by UAV swarms, this paper develops a multi-UAV collaborative search algorithm [7] based on distributed model predictive control, using coverage rate as the reward function. This algorithm seeks the optimal forward route for individual UAVs at each decision-making timestep, resulting in a collaborative search plan that meets Nash optimality. In a distributed architecture, the control of UAVs is relatively independent, enhancing the decision-making speed and disturbance resistance of the entire system [8]. Moreover, model predictive control helps to avoid redundant detection, thereby improving the search efficiency of the swarm.

To measure whether the exploration area has been explored, we have established a search graph model with uncertainty. To facilitate the description of the search status in a region while UAV swarms execute search missions, the area is gridded. Each grid cell $(i,j)$ is assigned a value $u_{ij}$ to represent the search status of the mission area. The map is discretized into $m\times n$ grid cells, and the uncertainty of each grid cell is defined as

$$u_{ij}=\begin{cases}1, & \text{Undetected}\\ 0, & \text{Detected}\\ -1, & \text{Obstacle}\end{cases}. \tag{3}$$

Authorized licensed use limited to: NANJING UNIVERSITY OF AERONAUTICS AND ASTRONAUTICS. Downloaded on January 05,2026 at 09:09:12 UTC from IEEE Xplore. Restrictions apply.

The initial uncertainty of all grid cells is set to 1. Once a grid cell $(i, j)$ falls within the detection range of a UAV, it is considered to be fully explored, and its uncertainty is reduced to 0. If an obstacle is detected in the grid cell, the uncertainty is set to -1. From this, the overall uncertainty $s$ of the entire mission area can be calculated, which is defined as

$$S = \sum_{i=1}^{m} \sum_{j=1}^{n} u_{ij}. \tag{4}$$

Each group of UAV Swarm performs relatively independent control and optimization decisions. They prioritize moving towards areas with the highest uncertainty while simultaneously avoiding obstacles. Based on the grid map, the direction of movement is divided into 8 directions. During the execution of their tasks, the UAV groups assess their flight paths in real time based on their current location and the status of the search graph, autonomously selecting the optimal direction to move. To evaluate the merits of various routes, a coverage reward function is defined as

$$J(k, \theta_i) = S(k+1) - S(k) \tag{5}$$

where $J(k, \theta_i)$ represents the reward for the UAV moving in direction $\theta_i$ at a time $k$. $S(k)$ is the map's uncertainty at the time $k$. $S(k+1)$ is the map's uncertainty at the time $k+1$ while $\theta_i$ is the target direction.

A larger coverage reward function indicates that the UAV's subsequent flight path can explore a larger area. To achieve optimal search efficiency, the path with the highest reward function is selected as

$$\theta^* = \arg\max_{i \in [1,d]} J(k, \theta_i) \tag{6}$$

where the direction angle $\theta^*$ represents the optimal movement direction for the UAV at time $k$.

In the later stages of detection, all surrounding areas of a UAV may have been fully explored, leading to situations where $J_{\max}(\theta) = 0$, meaning no reward can be gained from local movement and no optimal direction exists. In such cases, the UAV will resort to a global movement strategy.

In this global movement strategy, the map is then re-divided into 4×4 sections, and the sum of uncertainties for each section is calculated. The UAV selects the center of the section with the highest sum of uncertainties and the nearest distance as the globally optimal point. The UAV will then move towards this globally optimal point.

Due to the adoption of a distributed decision-making mechanism, potential path conflicts may arise among UAVs within the swarm. To address this, the study designs a negotiation-based decision-making process: each UAV, after making its decision, exchanges intentions with friendly UAVs within its communication range. The uncertainty of areas in the search graph detected by the intentions of all friendly UAVs is set to 0. Based on this, each UAV re-evaluates and decides its movement direction until the optimal movement state of all UAVs in the swarm satisfies Nash optimality. This approach realizes collaborative detection by a UAV swarm within a distributed architecture framework.

### *D. Collaborative Strike Strategy*

The swarm collaborative strike mission involves air-to-air combat between opposing UAV swarms, while also addressing the mission of striking enemy ground strongholds. Therefore, we have developed a distributed collaborative combat algorithm based on the relative distance and situational relationship between UAVs and targets [9]. By establishing a collaborative strike assessment function, we identified the optimal reachable position for an individual UAV within the next decision step. This approach facilitates the control of individual movements within the UAV swarm, thereby enabling the implementation of a distributed collaborative strike strategy based on the principle of proximity allocation. The use of distributed algorithm strategies enhances the swarm's robustness, effectively adapting to dynamically changing network structures.

The optimization goal of this distributed collaborative strike strategy is to maximize the collaborative strike assessment function, where the decision variables are the desired positions of each UAV in the swarm for their next move. Assuming the operational area is within the communication range of the swarm, each UAV can access fused perception information from the swarm. They can then proceed to their respective desired positions to execute corresponding combat actions. The collaborative combat actions that UAVs can perform include cooperation, confrontation, and Strike. "Cooperation" refers to UAVs working together, pooling their strength. "Confrontation" involves UAVs choosing to attack or retreat based on the current situation. "Strike" signifies UAVs conducting pinpoint strikes against the primary target. The evaluation function for the desired position concerning collaborative actions can be defined as

$$f_1 = \sum_{k \in \mathbb{N}} e^{-\frac{N_k^2}{2\sigma_1^2}} \tag{7}$$

where $\mathbb{N}$ represents the UAV around the desired position in the fused information. $N_k$ represents the distance between the desired position and UAV k. $\sigma_1$ is a parameter of the Gaussian function.

The evaluation function for the desired position concerning confrontation actions can be defined as

$$f_2 = \sum_{k \in \mathbb{Z}} S_k e^{-\frac{Z_k^2}{2\sigma_2^2}} \tag{8}$$

where $\mathbb{Z}$ represents the targets around the desired position in the fused information. $S_k$ represents the situation of the desired position relative to target k. $Z_k$ represents the distance between the desired position and target $k$. $\sigma_2$ is a parameter of the Gaussian function.

The evaluation function for the desired position concerning strike actions can be defined as

$$f_3 = e^{-\frac{c^2}{2\sigma_3^2}} \tag{9}$$

where $c$ represents the distance between the desired position and the primary target. $\sigma_3$ is a parameter of the Gaussian function.

Authorized licensed use limited to: NANJING UNIVERSITY OF AERONAUTICS AND ASTRONAUTICS. Downloaded on January 05,2026 at 09:09:12 UTC from IEEE Xplore. Restrictions apply.

Therefore, the maximum expected position evaluation function can be defined as

$$\max F = g(f_1 + f_2 + f_3) \tag{10}$$

where the position corresponding to $F$ in the equation represents the current UAV's desired location. The function $g(\cdot)$ adjusts the weights of $f_1$, $f_2$, $f_3$ based on the overall current UAV situation.

To solve the above models, the particle swarm optimization algorithm is adopted to obtain the optimal position. After obtaining the desired positions, each UAV adjusts its direction and speed for movement. During the confrontation process, UAVs only receive fused perception information from the swarm and no longer receive or follow any other instructions from the swarm.

### E. Effectiveness Evaluation Model

The intelligence of a UAV swarm system is primarily expressed through the emergence of collective intelligence, rather than simply stacking individual capabilities. Therefore, when assessing the effectiveness of UAV swarm warfare, it is crucial to focus on the swarm as a whole and conduct evaluations from a holistic perspective. The selection of metrics should be based on the capabilities of the swarm, rather than merely stacking the corresponding individual capabilities. In the construction of the performance indicators for UAV swarm intelligence, considering the fundamental characteristics of UAV swarm systems, a comprehensive assessment is made of the swarm's collaborative performance and operational efficiency. Based on the characteristics of swarm collaborative warfare tasks, the main evaluation indicators for UAV swarm warfare effectiveness are established, which include task completion rate, collaborative strike-to-loss ratio, and operational damage efficiency, as mentioned in reference [10]:

$$F = M_c + 1 - \exp(-M_{loss}) + T_r \tag{11}$$

where $M_c$ represents the task completion rate. $M_{loss}$ represents the collaborative strike-to-loss ratio. $T_r$ represents the swarm's operational damage efficiency. By considering the operational efficiency in the pre-mission, mid-mission, and post-mission phases, a comprehensive UAV swarm collaborative warfare effectiveness evaluation model is constructed. The weight coefficients for the pre-mission, mid-mission, and post-mission phases are respectively set as 0.2, 0.4, and 0.4.

## V. THE SIMULATION OF COOPERATIVE COMBAT OF UAVS

Based on typical scenarios for UAV swarm collaborative warfare, we have designed a series of combat scenarios for simulation verification and have evaluated the simulation results.

### A. Simulation System Realization

According to the requirements of the simulation experiments, this paper has designed a simulation system and a user interface based on Anylogic software. The user interface, enables interactions such as scenario loading, adjustment of UAV parameters, strategy selection, and real-time display of simulation scenarios and operational performance indicators in both two-dimensional and three-dimensional views, which is partly shown in Fig. 3. Scenario contents and environmental models are loaded into the simulation engine. Subsequently, each agent conducts the simulation based on loaded capability parameters, task procedures, and behavioral strategies. The simulation process can be observed in real time through a visualization interface.

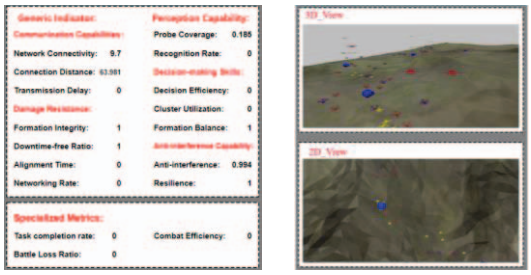



Fig. 3. Simulation Interface

### B. Scenario Design

The scenarios are designed for UAV swarm collaborative search and strike missions in the presence of unknown enemy targets and environments. The UAV swarm conducts cooperative searches and engages in combat tasks upon detecting enemy stronghold positions. The mission is considered complete when the enemy stronghold is destroyed. The setting of the scenario mission is:

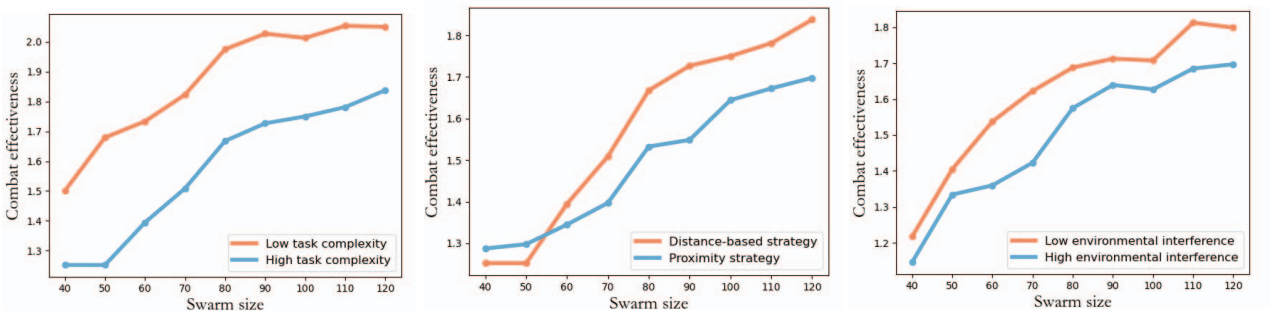
- Environment: Mountainous terrain with the presence of electromagnetic interference.
- Mission: Search enemy strongholds and carry out strikes.
- Our force composition: $m$ communication UAVs, $n$ detection UAVs, and $k$ strike UAVs, with a ratio of 1:4:5 among the three types of UAVs.
- Enemy force composition: $i$ strike UAVs and $j$ strongholds.

### C. Experimental Results and Analysis

Simulations were carried out under different scenario parameters. In our experiments, pertaining to the designed scenarios of swarm collaborative detection and collaborative strike missions, we conducted simulations under various scenarios with UAV swarm sizes ranging from 40 to 120. These experiments were tailored to situations involving both large-scale and small-scale enemy defense forces, aiming to explore the impact of swarm size, operational task complexity, environmental interference level, and swarm strategies on operational effectiveness, which are shown in Fig. 4.

A comparison was made between two different task complexities under different swarm sizes in Fig. 4(a), it can be observed that the effectiveness of swarm collaborative warfare is related to the swarm size. Within a certain range, as the swarm size increases, the operational effectiveness also increases. However, beyond a certain threshold, the impact of swarm size on operational effectiveness diminishes significantly. The operational effectiveness varies under different levels of task complexity, and the swarm size threshold varies as well. For lower task complexity, an 80 swarm size has reached the threshold, while for higher task complexity, the operational effectiveness continues to increase, with a threshold of no less than 120. And the combat effectiveness of swarm of the same scale under high task complexity is much lower than that under low combat complexity.

Authorized licensed use limited to: NANJING UNIVERSITY OF AERONAUTICS AND ASTRONAUTICS. Downloaded on January 05,2026 at 09:09:12 UTC from IEEE Xplore. Restrictions apply.




(a) (b) (c)

Fig. 4. Comparison of Simulation Effectiveness Evaluation

In Fig. 4(b), a comparison was made between two different collaborative strategies. The distance-based strategy is the collaborative strike strategy mentioned in Section IV, and the proximity strategy is a strategy based on heuristic rules. The shows that the effectiveness of two strategies is relatively similar under a limited swarm size. As the swarm size increasing, the effectiveness of Distance-based strategy outperforms significantly that of Proximity strategy in the current operational scenario. The Distance-based strategy can fully leverage the advantages of large-scale swarm collaborative operations, making combat effectiveness more sensitive to changes in swarm size.

In Fig. 4(c), in environmental models characterized by low environmental interference, the effectiveness of swarm combat operations surpasses that observed under conditions of high environmental interference. The difference is slight, which highlights the strong anti-interference properties of the cluster collaboration strategy adopted in this paper.

Based on the analysis of simulation experiments, the following decision-assisted conclusions can be drawn: The operational effectiveness of a swarm is positively correlated with its operational size within a certain range. The combat effectiveness of the Distance-based strategy is significantly better than that of the Proximity strategy, indicating that the former is more suitable for the current combat mission. Therefore, under combat missions characterized by high task complexity and high environmental interference level, it is advisable to opt for Distance-based strategy, with a swarm size selection within the range of 80 to 90. This approach facilitates the achievement of enhanced combat effectiveness and the generation of superior decision-making schemes, while concurrently optimizing for cost-efficiency in combat operations. Through comparative analysis of different strategies in simulation scenarios, this simulation system is capable to explore the optimal collaborative strategy and the best swarm force composition for specific operational tasks.

## VI. Conclusion

Regarding the issue of UAV swarm collaborative warfare, a simulation modeling design method has been designed in this paper. It includes the design of multiple behavioral model and strategy model for UAV swarms. Furthermore, a general system simulation model for UAV swarm collaborative warfare, suitable for verification through simulations, has been developed. An effectiveness evaluation model has also been designed to assess the performance of UAV collaborative warfare missions. The paper has specifically designed typical collaborative warfare scenarios and successfully implemented the full process of UAV swarm collaborative warfare in simulations. By conducting performance evaluations and analyzing simulation results, it enables functions such as combat effectiveness evaluation, comparison of various collaborative strategy algorithms, and guidance for decision optimization.

## References

[1] C. Wen, H. Wu, X. Wu, Y. Mao, H. Yin, “Air Combat Simulation System for Airborne Weapon Equipment Verification,” 2021 12th International Conference on Mechanical and Aerospace Engineering (ICMAE), Athens, Greece, 2021, pp. 467-470

[2] L. Duan, M. Wang, Z. Lin, J. Li, H. Shi, et al., “Research on Evaluation Method of UAV's Contribution Degree to Army Aviation Combat System Based on Simulation Deduction,” 2021 33rd Chinese Control and Decision Conference, Kunming, China, 2021, pp. 2580-2585.

[3] C. Liu, J. He, Q. Li, M. Wang, “An Air Combat Simulation System for Intelligent Decision-Making,” 2020 12th International Conference on Intelligent Human-Machine Systems and Cybernetics, Hangzhou, China, 2020, pp. 104-108.

[4] S. Qing and L. Fang, “Research on the Intelligent Combat Decision-Making under the Simulation and Deduction System,” 2021 International Conference on Big Data and Intelligent Decision Making, Guilin, China, 2021, pp. 206-209.

[5] Y. Feng, Y. Wu, H. Cao and J. Sun, “UAV Formation and Obstacle Avoidance Based on Improved APF,” 2018 10th International Conference on Modelling, Identification and Control, Guiyang, China, 2018, pp. 1-6.

[6] S. He, K. Shi, C. Liu, B. Guo, J. Chen, et al., “Collaborative Sensing in Internet of Things: A Comprehensive Survey, ” in IEEE Communications Surveys & Tutorials, vol. 24, no. 3, pp. 1435-1474, thirdquarter 2022.

[7] C. Wang, C. Chen, “A heuristic lowest unknown-degree target search strategy under non-structured environment for multi-agent systems”, Journal of Advanced Computational Intelligence and Intelligent Informatics, vol. 24, no. 7, pp. 934-943, 2020.

[8] N. Bo, X. Li, J. Dai and J. Tang, “A Hierarchical Optimization Strategy of Trajectory Planning for Multi-UAVs,” 2017 9th International Conference on Intelligent Human-Machine Systems and Cybernetics, Hangzhou, China, 2017, pp. 294-298.

[9] H. Liu, J. Zhang, P. Zu and M. Zhou, “Evolutionary Algorithm-Based Attack Strategy With Swarm Robots in Denied Environments,” in IEEE Transactions on Evolutionary Computation, vol. 27, no. 6, pp. 1562-1574, Dec. 2023.

[10] W. Quan, L. Jia, Z. Zhang, C. Chen and L. Wang, “A Multi-Dimensional Dynamic Evaluation Method for the Intelligence of Unmanned Aerial Vehicle Swarm,” 2023 IEEE International Conference on Unmanned Systems, Hefei, China, 2023, pp. 731-737.

Authorized licensed use limited to: NANJING UNIVERSITY OF AERONAUTICS AND ASTRONAUTICS. Downloaded on January 05,2026 at 09:09:12 UTC from IEEE Xplore. Restrictions apply.